

Ian Rossi Alvarado

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EDUCATION

Universidad Técnica Federico Santa María (UTFSM)
Ingeniería Civil Informática (Computer Science and Engineering)

2020 – Expected Nov. 2026

5.5-year professional engineering program (11 academic semesters), 300 ECTS-equivalent credits. All coursework completed; thesis defense pending.

Selected coursework: Introduction to Quantum Computing (95/100), Applied Cryptography Workshop (100/100), Security Systems (97/100), Scientific Computing (80/100), Computer Architecture and Organization, Computer Networks, Optimization, and Computational Statistics.

RESEARCH INTERESTS

Quantum computing and quantum error correction; machine learning and computer vision; scientific and high-performance computing; algorithms and systems.

RESEARCH EXPERIENCE

Undergraduate Thesis – Non-Destructive Estimation in Heavy ANFO: A CNN Regression Approach

2025 – 2026

Universidad Técnica Federico Santa María, Santiago, Chile

- Developed and evaluated a computer-vision regression system in Python using PyTorch, MobileNetV2, transfer learning, and OpenCV to estimate material composition from images.
- Trained the CNN on a 1,275-image dataset, achieving a mean absolute error of 4.92% on the independent test set and substantially outperforming a traditional computer-vision baseline.

PUBLICATIONS

Lucas Apablaza, Iván Oyarzún-Rojas, **Ian Rossi A.**, and Mauricio Solar. “Advances improved quantum genetic algorithms: methods, impact and future challenges.” *DYNA*, vol. 100, no. 6, pp. 491–496, 2025. DOI: 10.52152/D11434.

CONFERENCE PRESENTATIONS

Advances improved quantum genetic algorithms: methods, impact and future challenges

Oct. 29, 2025

2nd Quantum Computing Workshop, LI Latin American Computing Conference (CLEI 2025), Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile

- Presented the paper in an oral session as its sole presenter.

SELECTED ACADEMIC PROJECTS

BlastDynamics – Computer Vision Developer & Scrum Master

2025

Software Fair Project, Universidad Técnica Federico Santa María

- Sole developer of the video-analysis CLI for detecting and tracking flyrocks in blasting footage using frame differencing, morphological filtering, contour analysis, and trajectory association with OpenCV.
- Implemented pixel-to-metric trajectory estimation from camera parameters, distance, velocity, direction, and risk metrics, safety-perimeter calculation, annotated-video and JSON outputs, and batch processing integrated with the project API.

TEACHING EXPERIENCE

Teaching Assistant – Operating Systems

Universidad Técnica Federico Santa María, Santiago, Chile

Mar. 2025 – Jul. 2026

- Designed and graded laboratory assignments across three academic semesters.
- Provided technical guidance and supported students in reasoning through and debugging operating-systems coursework.

Teaching Assistant – User Interface Design

Universidad Técnica Federico Santa María, Santiago, Chile

Mar. 2025 – Jul. 2026

- Reviewed practical coursework in user-interface design across three academic semesters.
- Provided guidance on course material and student projects.

Teaching Assistant – Engineering, Informatics & Society

Universidad Técnica Federico Santa María, Santiago, Chile

Mar. 2023 – Dec. 2023

- Mentored student teams throughout the development of the course’s main semester project, providing guidance on planning, problem solving, and project execution.

TECHNICAL SKILLS

Programming	Python; PHP; JavaScript; academic experience with Go, C++, and Java.
ML & Computer Vision	PyTorch, OpenCV, CNNs, transfer learning, MobileNetV2.
Systems & Development	Linux/Unix, Git, GitHub, Docker.
Web & Platforms	Laravel, Harvard Dataverse (Java/XHTML), Apache Superset, HTML.
UX/UI	Figma, user interface design, user experience design, high-fidelity prototyping.

PROFESSIONAL EXPERIENCE

UX/UI Engineer & Developer

Chilean Institute for Disaster Resilience (Itrend), Santiago, Chile

Nov. 2025 – Present

Internship: Nov. 2025 – Feb. 2026 / Part-time: Mar. – Jul. 2026 / Full-time: Aug. 2026 – Present

- Serve as Itrend’s primary in-house UX/UI specialist while contributing to full-stack development across its digital platforms using Laravel/PHP and JavaScript.
- Led the redesign of the organization’s public website and developed product proposals and high-fidelity prototypes by translating client requirements into feasible product, UX, and technical specifications.

Software Development Intern

Agencia Ginebra, Santiago, Chile

Jan. 2025 – Mar. 2025

- Built and maintained WordPress websites, configured a Transbank payment integration, and served as the sole programmer on a small Godot-based game prototype.

LANGUAGES

Spanish: Native	English: Fluent (TOEFL iBT 110/120)	Japanese: Basic
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